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5

a Translation

b A translation of 3 units left and 8 units down.

c Congruence

6

a A reflection across the y-axis.

b Congruence

7

a A dilation by a factor of 0.6 using the origin as the center of dilation.

b Similarity

8

a Similar

b 2

9

a Similar

b 3

10

a Similar

b 3

11

a No **b** Yes **c** No **d** Yes

12

a No **b** No **c** No **d** Yes

e No **f** Yes

Additional questions

13

a A reflection across $\overleftrightarrow{EF}$

b $\overline{DE} \cong \overline{GE}$
$\overline{DF} \cong \overline{GF}$
$\overline{EF} \cong \overline{FE}$

c $\angle DEF \cong \angle GEF$
$\angle DFE \cong \angle GFE$
$\angle D \cong \angle G$

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14

a Triangles ABC and EDC are congruent. Triangle EDC is a rotation of triangle ABC by 180° around point C.

b Triangles PQR and SQR are congruent. Triangle SQR is a reflection of triangle PQR across $\overleftrightarrow{QR}$.

c Triangles UVW and XYZ are not congruent.

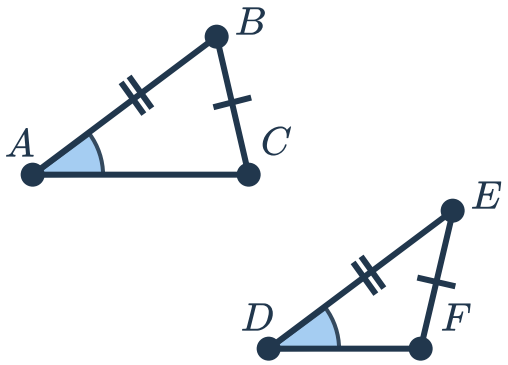
d Paralellograms $ABCD$ and $FGHJ$ are congruent. Parallelogram $FGHJ$ is a translation of parallelogram $ABCD$.

e Quadrilaterals $MNPO$ and $RQOP$ are not congruent.

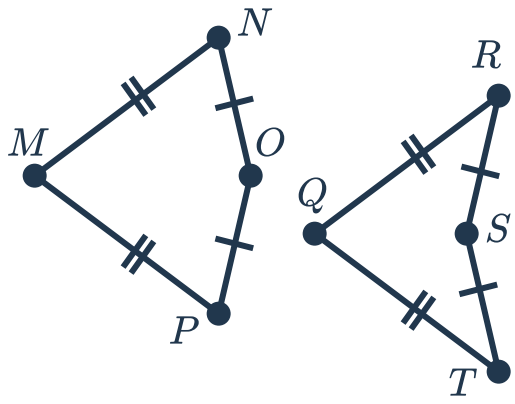
f Quadrilaterals $MJKL$ and $MNOP$ are congruent. $MNOP$ is a rotation of MJL around the point M.

15 These three types of transformations are all rigid transformations, meaning that they preserve segment lengths and angle measures. So if we use one or more of these transformations on a figure to produce a new figure, the new figure will have identical angle measures and side lengths to the original shape.

16



17



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18

a

i 2

ii Reflect across the y-axis and the dilate by a scale factor of 2

iii $(x, y) \to (-2x, -2y)$

b

i 3

ii Dilate by a scale factor of 3 then translate 6 units to the right, or translate 2 units to the right and then dilate by a scale factor of 3

iii $(x, y) \to (3x + 6, 3y + 6)$

c

i $\frac{5}{4}$

ii Answers may vary. Rotate 180° about the origin or reflect over the line $y = -x$ are two possible solutions.

iii $(x, y) \to (-2x, -2y)$

d

i $\frac{1}{2}$

ii Rotate 90° clockwise (or 270° counterclockwise) about the origin and dilate by a scale factor of $\frac{1}{2}$

iii $(x, y) \to (\frac{1}{2}y, -\frac{1}{2}x)$

19 The image of the tree is dilated by a scale factor of $\frac{1}{120}$, then reflected over the x-axis and translated the 20 ft between the eye and the object.

20

a True. The conditions for similarity are congruent angles and proportional sides. If two figures are congruent their angles are congruent and all corresponding sides will have a scale factor of 1 which makes them proportional.

b False. Rectangles and squares have congruent angles but are not similar.

c False

d True

21

a $(18, -18)$

b $(24, 24)$

c $(3, 3)$

d $(35, 39)$

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22

a Answers may vary.
Some considerations include the space between the edge of the screen and the caller's image and the actual size in pixels of the caller's image.

b Multiple assumptions must be made to accurately answer this question. Suppose the smaller image is $\frac{1}{8}$th the size of the entire screen. The image needs to be dilated by this scale factor using the origin as the center of dilation. This makes the dimensions of the smaller image 224×103 pixels as we can't have half a pixel. Then suppose there are about 25 pixels from both the top edge and side of the screen. That means we need to transate the image 703 pixels to the right and 1543 pixels up.

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